Progress Report 1

Project 1: Urban Drone Navigation and Workforce Scheduling Using AI

Course: COMP 569 Artificial Intelligence

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1. Project Overview

1.1 Project Topic

This project focuses on designing an AI system that integrates **informed search** and **constraint satisfaction** techniques to solve two interconnected real-world problems:

Optimal navigation of an autonomous drone in a 3D urban environment.

Scheduling qualified technicians to repair infrastructure tasks under constraints.

The system simulates a smart urban operations scenario where drones inspect city infrastructure and technicians are dispatched for necessary repairs.

1.2 Application Domain

The project falls under the domains of:

- Robotics and Autonomous Systems (drone navigation)
- Urban Infrastructure Management
- Operational Planning and Resource Allocation

These domains are highly relevant in modern smart cities, where autonomous aerial systems and optimized workforce scheduling improve efficiency, reduce operational costs, and enhance safety.

1.3 Motivation and Relevance

Autonomous drones are widely used for infrastructure inspection, emergency response, delivery services, and urban monitoring. Navigating complex 3D urban environments is challenging due to obstacles, no-fly zones, altitude changes, and limited energy resources, requiring intelligent path-planning strategies rather than simple shortest-path solutions.

Similarly, workforce scheduling is a critical optimization problem where technicians must be assigned to tasks based on skills, availability, and time constraints. This project demonstrates how AI techniques can efficiently address both spatial planning and resource allocation challenges in real-world urban systems.

2.Problem Formulation

Part A: Drone Navigation (Search Problem)

The drone navigation task is formulated as a state-space search problem defined by the tuple ⟨S, A, T, C, G⟩, where **S** represents the set of all possible drone states, **A** denotes the available actions, **T(s, a)** is the transition function that maps a state and action to a new state, **C(s, a, s′)** defines the cost of moving between states, and **G** represents the goal state. The objective of this problem is to compute an optimal path from a given start position to a goal position while minimizing total energy consumption.

Each state is represented as $s = (x, y, z)$, where x and y denote the horizontal coordinates and z represents the altitude. This three-dimensional representation captures realistic drone movement in an urban environment, including vertical changes required to avoid buildings and restricted zones.

The system takes as input a 3D grid environment, start and goal coordinates, obstacle locations, designated no-fly zones, and an energy-based cost model. The output consists of the optimal path from start to goal, the total energy cost of the path, and performance metrics such as the number of nodes expanded during search.

Several assumptions and constraints are applied to simplify modeling. The environment is discretized into a finite 3D grid, and movement actions include horizontal, upward, and downward transitions. All costs are non-negative, with horizontal and downward movements costing 1 unit and upward movement costing 2 units to reflect higher energy usage. Obstacles and no-fly zones are treated as impassable states. Battery limitations are modeled indirectly through cost minimization rather than explicit energy tracking.

This component of the project primarily involves search and planning. The environment is fully observable and deterministic, and the system does not incorporate learning or probabilistic inference.

Part B: Workforce Scheduling (Constraint Satisfaction Problem)

The workforce scheduling component is formulated as a **Constraint Satisfaction Problem (CSP)** defined by the tuple ⟨X, D, C⟩, where **X** represents the set of task variables, **D** represents the domains of possible values (available technicians), and **C** denotes the set of constraints that restrict valid assignments. The objective is to assign a technician to every task such that all constraints are satisfied simultaneously.

In this formulation, a state represents either a partial or complete assignment of technicians to tasks. The system takes as input a list of repair tasks, the required skills for each task, the skill sets of available technicians, and their availability schedules. The output is either a valid assignment of technicians to all tasks or a failure indication if no feasible solution exists.

The problem includes several constraints: technicians must possess the required skills for assigned tasks, must be available during the required time slots, must not be assigned to overlapping tasks, and each task must be assigned exactly one technician. Key assumptions include predefined task durations, static technician skill sets, and deterministic constraints. This component involves constraint satisfaction and combinatorial search, without incorporating learning or probabilistic inference.

3.Selected AI Approach

3.1 Informed Search for Navigation

The navigation component uses **Uniform Cost Search (UCS)** and the **A*** search algorithm. UCS serves as a baseline since it guarantees optimal solutions with non-negative costs, but it can be inefficient in large 3D environments. A* is selected as the primary approach because it improves efficiency by incorporating a heuristic while still guaranteeing optimality when the heuristic is admissible and consistent. The heuristics considered include Manhattan distance, Euclidean distance, and a building-aware heuristic that reflects obstacle height. These methods align with course concepts on informed search, heuristic design.

3.2 CSP Techniques for Scheduling

For workforce scheduling, the project employs backtracking search enhanced with forward checking, the Minimum Remaining Values (MRV) and degree heuristics, and Arc Consistency (AC-3). These techniques are well-suited because the problem involves finding valid assignments under explicit, deterministic constraints, and they efficiently reduce the search space. This approach aligns directly with course topics on constraint reasoning and search optimization.

4. System Design

The system consists of two main modules: the Drone Navigation Engine and the Workforce Scheduling Engine.

The **Drone Navigation Engine** includes an environment model, state generator, cost calculator, and search algorithm (UCS or A*). It takes the 3D environment, start and goal positions, and obstacles as input, generates possible states, calculates movement costs, and finds the optimal path. The workflow initializes the start state, expands nodes using a priority queue, applies heuristics, and stops when the goal is reached.

The **Workforce Scheduling Engine** includes a task manager, technician database, constraint checker, and backtracking solver. It takes tasks, technician skills, and availability as input and assigns technicians while enforcing constraints. The workflow selects tasks, assigns technicians, checks constraints, propagates effects, and backtracks if necessary, until a valid assignment is found.

Together, the modules integrate search and constraint reasoning to provide efficient solutions for drone navigation and workforce scheduling.

5. Expected Outcomes

The project will be successful if the drone navigation system finds an optimal, obstacle-free path with minimal energy cost and clearly shows that A* performs more efficiently than UCS. The workforce scheduling system must generate valid technician assignments that satisfy all constraints while demonstrating improved efficiency through heuristics. Performance will be evaluated using path cost, nodes expanded, and runtime for navigation, and backtracks, runtime, and domain reductions for the CSP component. Key challenges include designing an admissible building-aware heuristic, handling large 3D search spaces, managing tightly constrained schedules, and balancing realism with computational efficiency.

6. Progress Summary

As I am new to Artificial Intelligence and have intermediate Python skills, I am currently building my understanding of the core AI concepts required for this project, including state-space representation, UCS, A* search, heuristic design, admissibility, consistency, and CSP techniques such as backtracking, forward checking, MRV, and arc consistency. This foundation is helping me understand how the navigation and scheduling modules should be designed.

So far, I have completed the project definition and high-level system design. However, I still have basic questions such as how to select and validate an effective heuristic, how to properly compare A* and UCS performance, how grid size affects scalability, which CSP heuristics work best, and how to ensure all constraints are correctly enforced.

The remaining work includes full implementation of the search and CSP algorithms, conducting heuristic comparison experiments for the navigation module, performing CSP performance benchmarking, and completing detailed documentation and evaluation analysis.